

De-Anisotropizing Dense Embeddings with ABTT

Step 3 — Removing the parliamentary-register bias from sentence vectors

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1. Objective

Step 2 revealed that off-the-shelf dense embeddings of Knesset committee text are **severely anisotropic**: the cosine similarity between *any* two spans is ≈ 0.95 for **e5** and ≈ 0.98 for AlephBERT regardless of their gold topic, leaving a discrimination gap of only 0.007–0.018. This collapses clustering performance well below TF-IDF ($\text{ARI}_{\text{km}} = 0.377$ for TF-IDF vs. 0.322 for **e5**).

The hypothesis is that this isotropy is driven by a small number of directions shared across all committee discussions — the voice of a single Knesset committee’s procedural Hebrew (standard openings, motions, voting formulae). Projecting out these directions should widen the same-vs-different-topic margin and restore topic discriminability.

2. Method: All-But-The-Top (ABTT)

We apply the All-But-The-Top postprocessing introduced by Mu & Viswanath (ICLR 2018) (originally for static word vectors; applies equally to sentence embeddings):

1. **Center.** Subtract the mean of all 198 span vectors: $\hat{X} = X - \bar{x}$.
2. **Identify dominant directions.** Compute the economy SVD of $\hat{X}$; let $V \in \mathbb{R}^{k \times d}$ be the top- k right singular vectors.
3. **Project out.** Remove the subspace spanned by V : $\tilde{X} = \hat{X} - (\hat{X}V^\top)V$.
4. **Re-normalize.** L2-normalize each row of $\tilde{X}$.

We sweep $k \in \{0, 1, 2, 3, 5, 7, 10, 15, 20, 30, 50\}$ for each model (**e5**-large-1024d, **mpnet**-768d, **alephbert**-768d). $k = 0$ is the normalized baseline (same as Step 2, modulo centering). The PCA is fit on all 198 spans; evaluation is on the 105-span recurring subset.

3. Results

3.1 Quantitative summary

Configuration	Best k	ARI_{km}	ARI_{agg}	AUC	sim gap
TF-IDF (baseline)	—	0.377	0.249	0.741	0.069
e5 , $k = 0$ (raw)	—	0.322	0.136	0.744	0.018
e5 , $k = 1$	1	0.363	0.540	0.899	0.343
e5 , $k = 3$	3	0.375	0.442	0.861	0.259
mpnet , $k = 0$ (raw)	—	0.169	0.221	0.740	0.066
mpnet , $k = 10$	10	0.314	0.309	0.707	0.173
alephbert , $k = 0$ (raw)	—	0.224	0.113	0.632	0.007
alephbert , $k = 2$	2	0.391	0.513	0.887	0.344

AlephBERT $k = 2$ **beats TF-IDF** on KMeans ARI (0.391 vs. 0.377) — the first dense configuration to do so. **e5** $k = 1$ achieves the highest agglomerative ARI (0.540, a $2\times$ improvement over TF-IDF’s 0.249) and the highest AUC (0.899).

3.2 The anisotropy was concentrated in 1–2 principal components

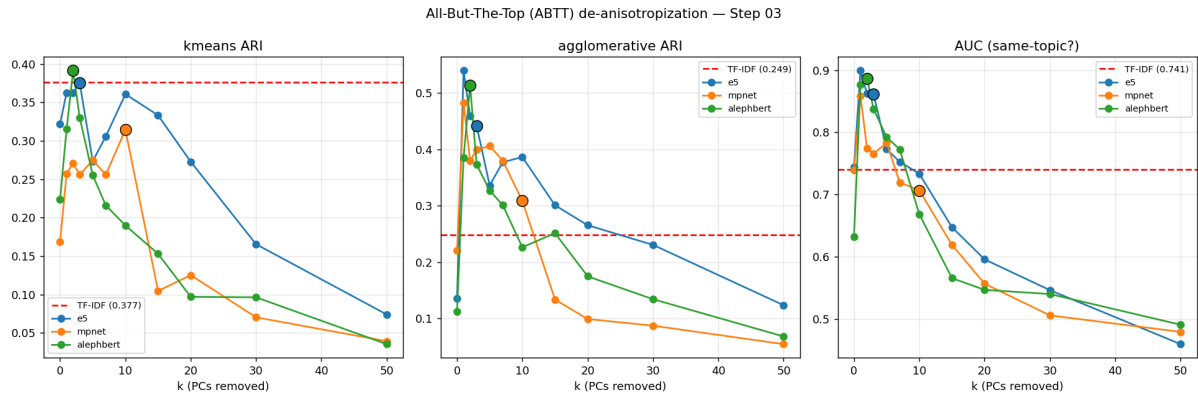
The similarity gap — the key diagnostic from Step 2 — improves dramatically with even a single component removed:

Model	gap, $k = 0$	gap, $k = 1$	gap, $k = 2$	gap, $k = 3$
e5	0.018	0.343	0.264	0.259
alephbert	0.007	0.333	0.344	0.252
TF-IDF	0.069 (constant)			

The first PC alone accounts for the bulk of the improvement. This is consistent with the hypothesis: a *single* “committee register” direction dominates all 198 span vectors, and removing it unlocks the topic-discriminative signal already latent in the encoder.

3.3 Over-removal degrades performance

Beyond $k \approx 5$, performance degrades monotonically as topic-discriminative directions are also projected out. The sweet spot of $k = 1-3$ is consistent across all three models.



KMeans ARI, Agglomerative ARI, and same-topic AUC as a function of k . The red dashed line is the TF-IDF baseline. Filled circles mark the best k per model.

4. Takeaways

- The anisotropy in Knesset committee embeddings is a **low-rank phenomenon**: $k = 2$ PCs suffice to encode almost all of the domain-register bias.
- **ABTT with $k = 2$ enables AlephBERT to surpass TF-IDF** — a result achievable in milliseconds of CPU postprocessing with no additional training.
- **e5 $k = 1$ is the AUC champion** (0.899 vs. 0.741 for TF-IDF), making it the natural choice for the similarity-threshold Task-2 baseline.
- The technique is *architecture-agnostic*: it works on any encoder whose output is dominated by a shared register subspace.

5. Future Directions

1. **Hybrid TF-IDF + ABTT score fusion.** TF-IDF contributes lexical precision (bill IDs, entity names) while ABTT-corrected dense contributes paraphrase recall. A weighted combination $\alpha \cdot S_{\text{tfidf}} + (1 - \alpha) \cdot S_{\text{dense}}$ should improve over either component alone. (Implemented in Step 4.)

2. **Finer sweep near $k = 1$ –2.** Fractional whitening (scale rather than zero-out singular directions) may give smoother trade-offs.
3. **Centroid-based representative in streaming.** Rather than using the first span as a journal-entry representative, maintain an ABTT-corrected centroid that updates as the entry accumulates spans.

Artifacts (this step): `outputs/abtt_results.json`, `outputs/abtt_curves.png`, `outputs/best_embeddings/{model}`

Code: `code/deanisotriize_eval.py`.